$$s'\sin\alpha = \frac{v_0'^2 \sin\alpha}{2\mu_2 g} = d \qquad (7)$$

Hence

$$\mu_2 = \frac{v_0'^2 \sin\alpha}{2gd} = 0.5\text{。} \qquad (8)$$

2. The block's friction force on the belt has magnitude

$$F = \mu_2 \cdot \frac{\eta g}{\mu_2 g} v_0' = \eta v_0', \qquad (9)$$

It points along v_0'. In the ground frame, belt speed is V. The block's work on the belt per unit time is

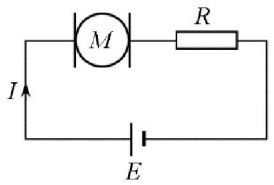
$$W = -FV\cos\alpha, \qquad (10)$$

Thus the additional power required by the load is

$$\Delta P = -W = \eta V^2 = 640\text{W}\text{。} \qquad (11)$$

The motor's no-load output power is

$$P_0 = I_0 E - I_0^2 R = 360\text{W}\text{。} \qquad (12)$$



Its output power with cargo is

$$P = P_0 + \Delta P = 1000\text{W}\text{。} \qquad (13)$$

Let I be the operating current with cargo. Then

$$P = IE - I^2 R, \qquad (14)$$

Solving gives

$$I = 10\text{A}\text{。} \qquad (15)$$

Scoring (8) (12) (13) (14) (15) total ; ; ; ;

2 (15) (18) (10) ; ; .

III. (20 points)

The light rod has length $2l$. At angle θ to the horizontal, ball 1, have speeds v_1, v_2. The rod's angular speed is ω. Conservation of mechanical energy gives

$$0 = mgl\sin\theta - 2mgl\sin\theta + \frac{1}{2}(2m)v_1^2 + \frac{1}{2}mv_2^2\text{。} \qquad (1)$$

Also,

$$v_1 = v_2 = l\omega, \qquad (2)$$

Solving (1)(2) gives

$$\omega = \sqrt{\frac{2g\sin\theta}{3l}} \qquad (3)$$